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HUJI Applicative Research Report — Week 33, 2026

8 high-potential paper(s) · Weekly · Week 33, 2026 · Generated August 10, 2026

This week's digest highlights significant advances across biomedical sciences, offering compelling opportunities for investors and industry partners. Featured innovations include breakthrough drug delivery platforms for genetic therapies, next-generation regenerative medicine for joint repair, and novel oncology therapeutics addressing drug resistance. We also present smart gene therapies for neurological disorders, a universal liquid biopsy diagnostic, and foundational tools for advanced materials and preclinical research. These technologies represent high-potential avenues for commercialization and patient impact.

RESEARCHERS & RESEARCH SUBJECTS — HIGH-POTENTIAL PAPERS

- **Eylon Yavin** — Drug Discovery, Genomics, Synthetic Biology
- **Amir Haze** — Medical Device, Clinical, Synthetic Biology, Proteomics
- **Yehuda Tzfati** — Drug Discovery, Genomics, Synthetic Biology, Clinical
- **Sheera Adar** — Drug Discovery, Clinical, Genomics
- **Mattan Hurevich** — Drug Discovery, Diagnostics, Materials, Synthetic Biology
- **Tawfeeq Shekh-Ahmad** — Drug Discovery, Neuroscience, Genomics, Clinical
- **Nir Friedman** — Diagnostics, Genomics, Clinical
- **Haitham Amal** — Drug Discovery, Neuroscience, Proteomics, Genomics

ALL BRANCHES — RESEARCH HIGHLIGHTS

1. Multiple Charged Purine (MCP) PNA as a Simple Mode for Cellular Uptake. 33/50

"Breakthrough PNA platform enables safe, simple intracellular genetic targeting without complex delivery."

Main researcher: Eylon Yavin

The Institute for Drug Research, The School of Pharmacy, The Faculty of Medicine, The Hebrew University of Jerusalem, Hadassah Ein-Kerem, Jerusalem 9112102, Israel.

Published in ACS polymers Au · 2026-06

ABOUT THIS RESEARCH

Researchers developed a new type of synthetic DNA analogue called MCP PNA that can naturally enter cells without needing complex delivery vehicles. This solves a major historical bottleneck where these stable molecules were too large or poorly charged to cross cell membranes.

COMMERCIAL ANGLE

Development of a versatile, high-stability platform for intracellular genetic targeting and antisense therapeutics without the need for expensive lipid nanoparticles or viral vectors.

COMMERCIALISATION METRICS

Novelty 6/10

The work provides a practical chemical modification to improve PNA uptake, but utilizing charge modification for cellular delivery is a known strategy in oligonucleotide chemistry rather than a groundbreaking platform shift.

Commercial Potential 8/10

This represents a platform-enabling technology that solves a fundamental bottleneck (cellular uptake) for PNA, significantly expanding the addressable market for PNA-based therapeutics and diagnostics.

Market Size 9/10

The work addresses a fundamental delivery bottleneck for PNA, a platform with massive potential across antisense therapy, diagnostics, and gene editing markets. Solving cellular uptake enables a wide range of therapeutic applications, making this a platform-enabling advancement with a very large total addressable market.

Tech Readiness 3/10

The work focuses on fundamental chemical modification of PNA for cellular uptake, representing basic proof-of-concept laboratory research without evidence of in vivo efficacy or a developed delivery platform.

IP Strength 7/10

The introduction of a specific chemical modification (MCP) to solve a fundamental PNA delivery bottleneck is highly patentable as a composition of matter. While it enables a platform for cellular uptake, its ultimate strength depends on the breadth of the claims regarding the purine modifications.

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2. Recombinant Human Amelogenin Protein Enhances Healing of Osteochondral Injury in a Goat Model. 33/50

"Regenerative protein for joint repair shows significant healing of cartilage and bone in goats."

Main researcher: Amir Haze

Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem, Israel.; Orthopedic Surgery Department, Hadassah-Hebrew University Medical Center, Jerusalem, Israel.; The Wohl Institute for Translational Medicine, Hadassah-Hebrew University Medical Center, Jerusalem, Israel.

Published in Journal of orthopaedic research : official publication of the Orthopaedic Research Society · 2026-04

ABOUT THIS RESEARCH

Researchers tested a recombinant human amelogenin protein product called Remelix® in a goat model to treat cartilage and bone injuries. The study showed that the protein significantly promotes the healing of both cartilage and subchondral bone without adverse effects.

COMMERCIAL ANGLE

Development of a high-value regenerative medicine biologic for treating osteoarthritis and traumatic joint injuries in humans.

COMMERCIALISATION METRICS

Novelty 4/10

The research focuses on validating a known biological mechanism (amelogenin) in a larger animal model rather than introducing a new molecular entity or novel platform. While the transition from rat to goat is a critical translational step, it represents incremental refinement of existing prior art rather than a groundbreaking scientific innovation.

Commercial Potential 8/10

The transition from rodent models to a large animal (goat) model significantly de-risks the technology and demonstrates clear readiness for clinical translation. The mention of a specific, branded product (Remelix®) and a defined carrier (PX 407) suggests a high degree of developmental maturity and a clear path toward a regulated medical device or biologic.

Market Size 8/10

The target market for osteochondral repair is substantial, driven by a high prevalence of joint injuries and the significant clinical unmet need for regenerative solutions in an aging population. The transition from rat to large animal models suggests a high potential for clinical translation into a large, high-value orthopedic market.

Tech Readiness 6/10

The study has progressed from small animal models to a large animal (goat) model, which is a critical regulatory milestone for orthopedic biologics. However, it remains in the preclinical stage and has not yet entered human clinical trials.

IP Strength 7/10

The use of a branded recombinant protein (Remelix®) suggests established manufacturing processes and likely composition-of-matter or process patents. However, the reliance on a well-known protein like amelogenin may face challenges regarding non-obviousness compared to existing recombinant growth factor technologies.

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3. Addendum: Telomouse-a mouse model with human-length telomeres generated by a single amino acid change in RTEL1.

32/50

"Novel mouse model with human-length telomeres offers a superior platform for anti-aging drug discovery."

Main researcher: Yehuda Tzfati · tzfati@mail.huji.ac.il.

Department of Genetics, The Silberman Institute of Life Sciences, Safra Campus, The Hebrew University of Jerusalem, Jerusalem, 91904, Israel. tzfati@mail.huji.ac.il.

Published in Nature communications · 2026-05

ABOUT THIS RESEARCH

Researchers have developed a specialized mouse model that possesses human-like telomere lengths through a precise single amino acid modification in the RTEL1 protein. This model allows for more accurate preclinical testing of aging-related diseases and therapies by better mimicking human cellular biology.

COMMERCIAL ANGLE

The creation of a high-fidelity biological surrogate for human aging enables pharmaceutical companies to accelerate the development and validation of anti-aging and longevity drugs.

COMMERCIALISATION METRICS

Novelty 8/10

The research demonstrates a highly innovative genetic engineering approach by using a single amino acid substitution to recapitulate a complex human phenotype, creating a powerful, platform-enabling model for aging and telomere biology.

Commercial Potential 6/10

While the creation of a humanized telomere mouse model is a valuable tool for aging and oncology research, its primary value is as a specialized research reagent rather than a broad-market therapeutic or platform technology.

Market Size 7/10

A mouse model with human-length telomeres serves as a high-value platform for aging and oncology research, potentially tapping into the massive biologics and longevity drug discovery markets. However, as a specialized research tool, its market is primarily limited to academic and pharmaceutical R&D budgets rather than direct clinical application.

Tech Readiness 3/10

The work describes the creation of a fundamental biological research tool (a transgenic mouse model) rather than a therapeutic or diagnostic product. While it enables downstream drug discovery, the technology itself is at the basic research/proof-of-concept stage for modeling human telomere biology.

IP Strength 8/10

The creation of a transgenic animal model via a specific single amino acid mutation in a highly conserved gene (RTEL1) represents a platform-enabling tool with strong potential for patentability regarding the specific genetic construct and its utility in humanized telomere research.

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4. A Cisplatin-Based Prodrug Inhibits Nucleotide Excision Repair Independently of Chromatin Accessibility to Overcome Resistance. 32/50

"New platinum prodrug effectively overcomes cisplatin resistance, a major challenge in cancer therapy."

Main researcher: Sheera Adar

Department of Microbiology and Molecular Genetics, The Institute for Medical Research Israel-Canada, The Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem 9112102, Israel.

Published · 2026-04

ABOUT THIS RESEARCH

Researchers have developed a new platinum-based drug derivative that combines cisplatin with two inhibitors to combat chemotherapy resistance. The study reveals that this new drug works by increasing drug uptake and blocking DNA repair mechanisms rather than just opening up chromatin.

COMMERCIAL ANGLE

The development of multifunctional platinum(IV) prodrugs offers a high-value therapeutic strategy to overcome cisplatin resistance in oncology patients.

COMMERCIALISATION METRICS

Novelty 7/10

The research introduces a novel Pt(IV) prodrug scaffold that mechanistically diverges from the conventional 'chromatin accessibility' hypothesis for KDAC combinations. While the chemical modification of cisplatin is a known strategy, the specific discovery of NER inhibition as the primary driver offers a new therapeutic logic for overcoming resistance.

Commercial Potential 7/10

The development of a novel Pt(IV) prodrug (cPVP) offers a clear path toward a single-agent therapeutic that addresses complex cisplatin resistance, representing a high-value drug candidate. While the mechanism is well-defined, the commercial potential is slightly tempered by the inherent competition with existing combination therapies and the complexity of optimizing a multi-ligand Pt(IV) scaffold for clinical safety.

Market Size 8/10

Cisplatin is a foundational, high-volume chemotherapy used across numerous cancer indications, meaning any effective resistance-overcoming derivative targets a massive, multi-billion dollar global oncology market. The ability to bypass complex resistance mechanisms like NER provides a high-value clinical solution for a substantial patient population.

Tech Readiness 3/10

The research is currently at the lead discovery/in vitro stage, focusing on the synthesis and mechanistic validation of a novel prodrug in a laboratory setting. While it identifies a promising mechanism to overcome resistance, it lacks the preclinical animal models or pharmacokinetic data required to move toward clinical development.

IP Strength 7/10

The synthesis of a novel Pt(IV) prodrug (cPVP) represents a clear opportunity for composition-of-matter patent protection. While the combination approach is highly defensible, the score is tempered by the fact that the individual components (valproic acid and phenylbutyrate) are known entities, potentially limiting the breadth of the intellectual property moat.

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5. Total Synthesis of Conjugation-Ready Sulfated Red Algae Carrageenan Oligosaccharides for Sensing Applications.

31/50

"Precision synthesis of complex carrageenan building blocks unlocks advanced drug delivery and biosensing."

Main researcher: Mattan Hurevich

Institute of Chemistry and Center for Nanoscience and Nanotechnology, The Hebrew University of Jerusalem, Safra Campus, Givat Ram, Jerusalem 9190401, Israel.

Published in Journal of the American Chemical Society · 2026-06

ABOUT THIS RESEARCH

Researchers have developed a new method to chemically synthesize precise, uniform fragments of carrageenan, a complex seaweed polysaccharide that is usually too irregular to control. This allows for the creation of customized oligosaccharides with specific sulfation patterns and attachment points for specialized use.

COMMERCIAL ANGLE

The ability to manufacture highly controlled, homogeneous carrageenan building blocks enables the development of advanced targeted drug delivery systems and highly sensitive biosensors.

COMMERCIALISATION METRICS

Novelty **8/10**

The work represents a significant leap in glycan engineering by achieving the first total synthesis of homogeneous carrageenan oligogalactans, overcoming long-standing structural complexity barriers. This provides a platform-enabling methodology for precision sulfated polysaccharides that was previously inaccessible via natural extraction.

Commercial Potential **6/10**

The paper provides a high-value chemical platform for creating precise, functionalized building blocks, but the commercial application is currently limited to niche sensing and R&D tools rather than a large-scale industrial material.

Market Size **7/10**

The ability to produce homogeneous, precisely sulfated carrageenan oligosaccharides unlocks high-value niche markets in precision drug delivery and advanced diagnostic sensing. However, while the platform is enabling, the total addressable market is currently constrained by the specialized, high-margin nature of synthetic glycans compared to bulk commodity carrageenans.

Tech Readiness **3/10**

The work is at a fundamental academic stage, focusing on the complex chemical proof-of-concept for total synthesis rather than scalable manufacturing or functional application testing. While the 'conjugation-ready' nature provides a pathway for future platforms, the technology remains in the basic research/lab-scale phase.

IP Strength **7/10**

The development of a specialized, multi-level protecting group hierarchy for total synthesis of complex polysaccharides represents a high barrier to entry and strong potential for composition-of-matter patents. While the synthetic methodology is highly defensible, its commercial impact depends on the successful application of these oligosaccharides in specific high-value sensing or therapeutic markets.

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6. Activity-dependent antioxidant gene therapy protects against epileptogenesis. 31/50

"Smart gene therapy prevents epilepsy by activating antioxidant enzymes only during brain activity."

Main researcher: Tawfeeq Shekh-Ahmad · Tawfeeq.Shekh-Ahmad@mail.huji.ac.il.

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Published in Signal transduction and targeted therapy · 2026-06

ABOUT THIS RESEARCH

This research describes a gene therapy approach that uses natural brain activity to trigger the production of antioxidant enzymes. This mechanism aims to prevent the development of epilepsy following an initial brain injury.

COMMERCIAL ANGLE

Development of smart, responsive gene therapies for neurological disorders that activate only during specific physiological states to minimize side effects.

COMMERCIALISATION METRICS

Novelty 7/10

The work is highly innovative for shifting the therapeutic paradigm from reactive seizure control to proactive, activity-dependent gene modulation to prevent epileptogenesis. However, while the mechanism is sophisticated, antioxidant gene therapy is an established concept, making the novelty incremental rather than a complete disruption of biological principles.

Commercial Potential 7/10

The technology targets a high-unmet-need clinical indication (epileptogenesis) with a specific mechanism (antioxidant gene therapy), offering a clear path toward a specialized therapeutic product. However, the lack of abstract details regarding delivery vectors and scalability limits a higher score.

Market Size 7/10

The target market for epilepsy therapeutics is massive and high-value, representing a multi-billion dollar global neurology segment. However, the score is tempered by the specific niche of epileptogenesis prevention, which is a subset of the broader epilepsy market and faces complex clinical adoption hurdles.

Tech Readiness 3/10

The title suggests a fundamental mechanistic study using gene therapy in a disease model, which typically implies preclinical, lab-based proof-of-concept research rather than a developed therapeutic candidate.

IP Strength 7/10

The use of activity-dependent promoters for targeted gene therapy provides a strong basis for composition-of-matter and method-of-use patents, though defensibility depends on the novelty of the specific promoter/transgene combination.

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7. Using Disease-Agnostic Genomic Liquid Biopsy in Complex Diagnoses: Real-Life Example of cfChIP-Seq in a Fever of Unknown Origin.

30/50

"Universal liquid biopsy platform rapidly diagnoses complex, rare diseases without prior hypotheses."

Main researcher: Nir Friedman

The Lautenberg Center for Immunology and Cancer Research, Faculty of Medicine, and the Rachel and Selim Benin School of Computer Science and Engineering, The Hebrew University of Jerusalem, Jerusalem, Israel.; The Hebrew University Center for Computational Medicine, The Hebrew University of Jerusalem, Jerusalem, Israel.

Published · 2026 Aug 7

ABOUT THIS RESEARCH

Researchers utilized a specialized sequencing technique called cfChIP-Seq to analyze cell-free DNA in the blood to diagnose a complex case of fever of unknown origin. This approach allows for a disease-agnostic search for genomic signatures of inflammation or infection without needing a prior hypothesis.

COMMERCIAL ANGLE

Development of a universal liquid biopsy diagnostic platform for rapid identification of rare or occult diseases in critical care settings.

COMMERCIALISATION METRICS

Novelty 7/10

The application of cfChIP-Seq as a disease-agnostic diagnostic tool for Fever of Unknown Origin represents a novel translation of epigenetic profiling to clinical diagnostics, moving beyond traditional oncology uses. However, the novelty is tempered by the fact that it applies existing ChIP-Seq methodology to a new clinical indication rather than introducing a fundamentally new molecular biology technique.

Commercial Potential 7/10

The use of cfChIP-Seq as a disease-agnostic diagnostic platform for complex cases like FUO suggests high versatility and a clear path toward a high-value clinical service. However, commercial scaling is tempered by the current complexity of ChIP-Seq workflows compared to standard NGS.

Market Size 8/10

The technology is a disease-agnostic platform for complex diagnoses, meaning its addressable market extends beyond a single indication to all undiagnosed systemic illnesses and infectious diseases. This platform-enabling approach significantly maximizes potential market size compared to targeted assays.

Tech Readiness 4/10

The paper presents a 'real-life example' case study, indicating the technology is functional in a clinical setting, but it remains a research-grade application of cfChIP-Seq rather than a standardized, scalable diagnostic product.

IP Strength 4/10

The paper describes the application of cfChIP-Seq, an established genomic method, to a specific clinical use case (FUO) rather than introducing a novel, patentable platform or proprietary molecular tool. Defensibility relies on clinical validation and data rather than a protectable technological breakthrough.

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8. Phosphoproteomic profiling reveals reversal of dysregulated kinase signaling by nitric oxide inhibition in the Shank3 mouse model of autism. 28/50

"New molecular targets identified for precision pharmacological interventions in autism spectrum disorders."

Main researcher: Haitham Amal · Haitham.amal@mail.huji.ac.il.

Institute for Drug Research, School of Pharmacy, Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem, Israel. Haitham.amal@mail.huji.ac.il.; The Rosamund Stone Zander and Hansjoerg Wyss Translational Neuroscience Center, Boston Children's Hospital, Boston, MA, USA. Haitham.amal@mail.huji.ac.il.

Published in Scientific reports · 2026-06

ABOUT THIS RESEARCH

Researchers used phosphoproteomic profiling to map how nitric oxide inhibition restores normal signaling pathways in a mouse model of autism related to the SHANK3 gene. The study identifies specific protein phosphorylation changes that could explain how targeting nitric oxide synthase can reverse behavioral and synaptic symptoms.

COMMERCIAL ANGLE

The findings provide high-resolution molecular targets for the development of precision pharmacological interventions aimed at correcting dysregulated kinase signaling in autism spectrum disorders.

COMMERCIALISATION METRICS

Novelty 6/10

The study provides meaningful mechanistic insight by mapping the signaling pathways downstream of NO inhibition, but it follows a predictable biological logic rather than introducing a groundbreaking new paradigm or platform technology.

Commercial Potential 6/10

The study identifies a mechanistic target (nNOS inhibition) for a high-unmet-need indication, but the phosphoproteomic focus is primarily descriptive rather than proposing a novel, druggable modality or proprietary platform.

Market Size 9/10

Autism spectrum disorder represents a massive, global therapeutic market with significant unmet medical need and a high prevalence across diverse demographics. Targeting a fundamental neurobiological pathway like nNOS/phosphoproteomics offers potential for high-value, broad-spectrum clinical application.

Tech Readiness 3/10

The research is currently at the fundamental discovery stage, utilizing animal models (Shank3 mice) to identify mechanistic signaling pathways rather than testing a clinical candidate in humans. While it provides high-value biological validation, it lacks the preclinical development or pharmacological optimization required for higher TRL scores.

IP Strength 4/10

The paper describes fundamental mechanism-of-action (MoA) discovery rather than a novel composition of matter; while identifying signaling pathways is valuable for target validation, the use of nitric oxide inhibition is an established biological phenomenon that offers limited new patentable space.

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